Yu Chul (Jerry) Lee

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EDUCATION

The University of Texas at Austin
Phd in Mechanical Engineering — Expected Graduation: May 2028
Master of Science in Mechanical Engineering — Graduation: Dec 2024
GPA: 4.0/4.0

Yonsei University
Bachelor of Science in Mechanical Engineering — Graduation: Aug 2023
GPA: 3.71/4.5

RESEARCH EXPERIENCE

Graduate Research Assistant
Mobility Systems Laboratory (Prof. Junmin Wang), The University of Texas at Austin — Jan 2025 – Present

Interaction and Mode-Aware Intelligent Driver Model

- Developed a hybrid physics-informed machine learning architecture for longitudinal driver behavior prediction at four-way intersections.
- Designed an interaction-aware and driver-mode-aware Intelligent Driver Model (IDM) augmented with learned latent representations for realistic acceleration prediction.
- Trained and evaluated models using **163 NGSIM intersection scenarios** and human-driver simulator datasets.
- Achieved **0.623 m/s velocity RMSE** over a 2-second prediction horizon.
- Demonstrated strong generalization to unseen scenarios, reducing acceleration prediction RMSE from **20.5 m/s² to 4.06 m/s²** compared with Explainable Boosting Machine (EBM) baselines.
- Developed data-processing and evaluation pipelines in MATLAB and Python for scenario extraction, feature engineering, model training, and rollout analysis.

Explainable Driver Modeling and Personalization

- Developed an interpretable driver-modeling framework using SHAP-based feature selection and Explainable Boosting Machines (EBMs).
- Engineered physically meaningful driver-behavior features from vehicle kinematics and traffic interactions.
- Proposed a residual-learning adaptation method enabling personalized driver modeling from sparse human-driving datasets.
- Achieved **94.4% fidelity** when approximating synthetic driver models and up to **88.5% fidelity** for human-driver behavior.
- Demonstrated preservation of individual driving styles, achieving **75% driver identification accuracy** on unseen driving data.

Engineering EXPERIENCE

EcoCAR: Connected and Autonomous Vehicle Team
Team Leader — Jan 2025 – May 2026

- Lead a multidisciplinary team of **8 undergraduate and graduate students** developing connected and automated vehicle (CAV) technologies on a Cadillac LYRIQ platform.
- Directed development, validation, and deployment of Cooperative Adaptive Cruise Control (CACC), Lane Keeping Assist (LKA), Automated Intersection Navigation (AIN), sensor-fusion, and V2X-enabled autonomy features.
- Coordinated model-based development workflows using MATLAB/Simulink, RTMaps, dSPACE ASM, and dSPACE AUTERA.

Cooperative Adaptive Cruise Control (CACC)

- Co-developed a Model Predictive Control (MPC) based CACC system utilizing V2V communication in SAE J2735
- Integrated vehicle position and velocity information from V2X radios into real-time control algorithms.
- Implemented CAN-FD communication between dSPACE AUTERA and production vehicle control systems.
- Successfully deployed and validated the controller on a Cadillac LYRIQ during closed-course testing.
- Achieved vehicle-level performance targets including: (**0.062 m steady-state spacing RMSE| 2.999 m/s³ maximum jerk| 1.707 m/s² maximum acceleration**)

Lane Keeping Assist (LKA)

- Guided development of a Sliding Mode Controller (SMC) for lane-keeping using LiDAR and camera perception inputs.
- Mentored undergraduate team members in controller design and validation.
- Validated controller performance through HIL and vehicle testing.
- Achieved: **0.121 m lateral tracking error, 0.35 m/s³ lateral jerk** in HIL testing
- Conducted 5 vehicle-in-the-loop experiments and identified sensor-latency limitations impacting closed-loop stability.

SELECTED PROJECTS

Autonomous Vehicle Project

Team Leader — Jan 2024 – December 2024

- Designed and tested vehicle path tracking controller to achieve lane change at 40 m/s with 0.1 m lateral error
- Performed analytical calculations with numerical analysis using bode plot and root locus to meet system requirements
- Reduced lateral tracking RMSE by: (**52.98%** single lane changes| **48.61%** double lane changes)
- Achieved performance comparable to MPC while requiring significantly lower computational effort.
- Published as first author in IEEJ Journal of Industry Applications.

Advanced Mechatronics Competition (1st place)

Team Member — August 2023 - December 2023

- Designed and manufactured an autonomous mobile robot capable of target localization and pneumatic actuation
- Manufactured robot base and 12 custom parts with CAD(Solidworks) to fit within space limit of 1 foot by 1 foot
- Tested 5 prototypes of the feeding and shooting mechanism and was able to shoot 1.5m without jamming mechanism

PUBLICATIONS

Controls & Vehicle Dynamics: Model Predictive Control (MPC), Sliding Mode Control, LQR, Feedback linearization
Programming: MATLAB, Simulink, Python, C
Autonomous Systems: V2X Communication (SAE J2735), CAN-FD, Driver Modeling, Trajectory Prediction
Simulation & Deployment: dSPACE AUTERA, dSPACE ASM, RTMaps, Unreal Engine, ROS, CANalyzer, HIL, VIL
Machine Learning & Data Analysis: Explainable Boosting Machines (EBM), SHAP, Random Forests, Clustering
Certifications: Fundamentals of Engineering (FE Mechanical), Passed 2023

PUBLICATIONS

Yu Chul Lee, Chi-Chang Huang, Umesh Malepati, Xingyu Zhou, and Junmin Wang, "Practical Method for High-Speed Path-Tracking Control of Vehicles with Time-Delayed Steering System Dynamics," IEEJ Journal of Industry Applications, vol. 15, no. 1, 2025.

Yu Chul Lee and Junmin Wang, "Human Impact of Visual Detail in Vehicle Lane-Keeping System Communication," Proceedings of 2025 IFAC Symposium on Intelligent Autonomous Vehicles, May 2025.